SRM Institute of Science and Technology
College of Engineering and Technology
School of Computing
(Common to all Branches)
SRM Nagar, Kattankulathur – 603203, Chengalpattu District, Tamilnadu

Mode of Exam
OFFLINE

Academic Year: 2023-24 (ODD)

RegNo

Test: CLAT-2 **Date: 09-10-2023**
Course Code & Title: 21CSS201T COA **Duration: 100 minutes**
Year & Sem: II & III **SET B** **Max. Marks: 50**

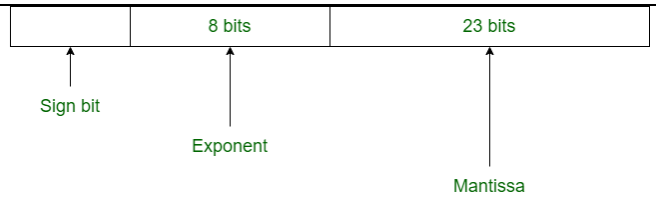
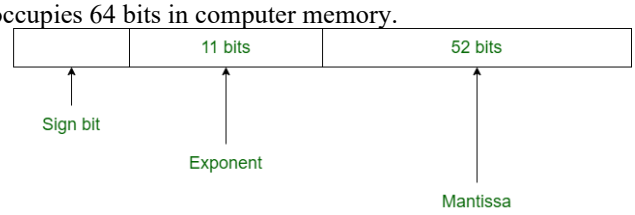
Course Articulation Matrix:

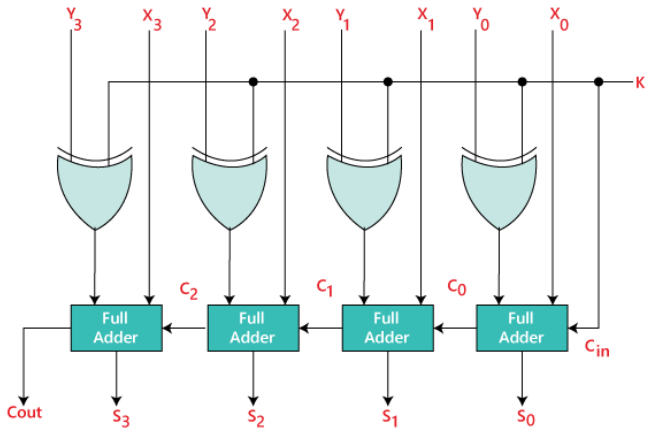
Course Learning Outcomes (CLO)	At the end of this course, learners will be able to:	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO-2	Apply Boolean algebra as related to designing computer logic, through simple combinational and sequential logic circuits	H	M	-		-	-	-	-	-	-	-	-
CO-3	Examine the detailed operation of Basic Processing units and the performance of Pipelining	H	-	-		-	-	-	-	-	-	-	-

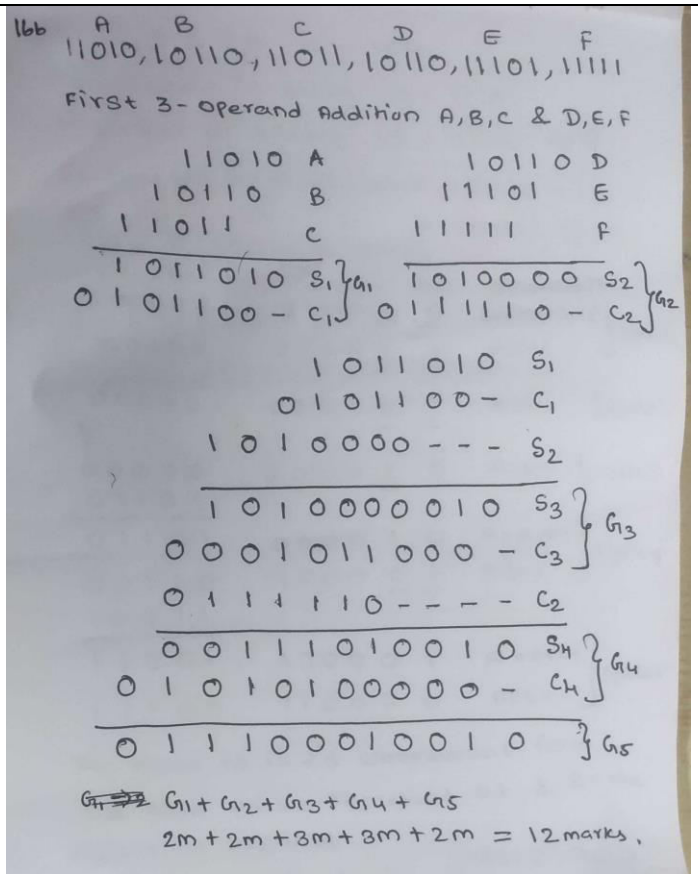
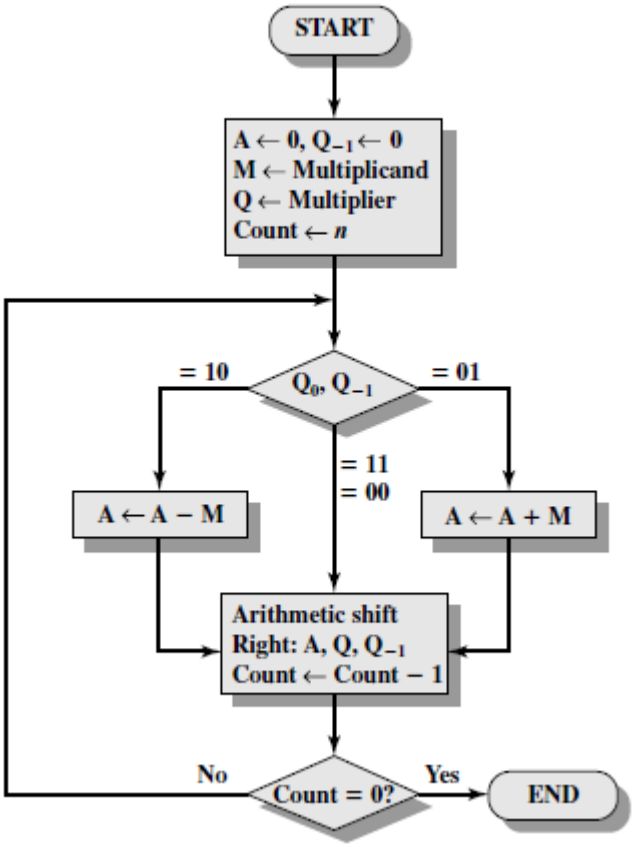
Part – A
(1 x 10 = 10 Marks)
Instructions: Answer all

Q. No	Questions	Marks	BL	CO	PO	PI Code
1	When using Branching, the usual sequencing of the PC is altered. A new instruction is loaded which is called as _____ a) Branch target b) Loop target c) Forward target d) Jump instruction	1	1	2	1	2.1.2
2	In the following indexed addressing mode instruction, MOV 5(R1), LOC the effective address is _____ a) EA = 5+R1 b) EA = R1 c) EA = [R1] d) EA = 5+[R1]	1	2	2	1	2.1.2
3	The microprocessor of a computer can operate on any information if it is present in _____ only. a) Program Counter b) Flag c) Main Memory d) Secondary Memory	1	1	2	1	2.1.2
4	The assembler directives which are the hints using some predefined alphabetical strings are given to a) processor b) memory c) assembler d) processor & assembler	1	1	2	1	2.1.2

5	**The method used to reduce the maximum number of summands by half is _____** **a) Fast multiplication** **b) Bit-pair recording** **c) Quick multiplication** **d) Booth Multiplication**	**1**	**1**	**3**	**1**	**2.1.2**
6	**In IEEE 32-bit representations, the mantissa of the fraction is said to occupy _____ bits.** **a) 24** **b) 23** **c) 20** **d) 16**	**1**	**1**	**3**	**1**	**2.1.2**
7	**Half-adders have a major limitation, they cannot _____** **a) Accept a carry bit from a present stage** **b) Accept a carry bit from a next stage** **c) Accept a carry bit from a previous stage** **d) Accept a carry bit from the following stages**	**1**	**1**	**3**	**1**	**2.1.2**
8	**Which of the following is used for binary multiplication?** **a) Restoring Multiplication** **b) Booth's Algorithm** **c) Pascal's Rule** **d) Digit-by-digit multiplication**	**1**	**1**	**3**	**1**	**2.1.2**
9	**DeMorgan's theorem states that _____** **a) (AB)' = A' + B'** **b) (A + B)' = A' * B** **c) A' + B' = A'B'** **d) (AB)' = A' + B**	**1**	**2**	**3**	**1**	**2.1.2**
10	**The product of 1101 & 1011 is _____** **a) 10001111** **b) 10101010** **c) 11110000** **d) 11001100**	**1**	**1**	**3**	**1**	**2.1.2**
	Part – B **(4*4 = 16 Marks)** **Answer any four questions**					
11	**Write a program to evaluate arithmetic expression using stack organized computer with 0-address instructions.** **X= (A-B) * (((C-D * E) /F) /G)** Post Fix Form X = AB-CDE*-F/G/* PUSH A TOS <- A PUSH B TOS <- B SUB TOS <- (A-B) PUSH C TOS <- C PUSH D TOS <- D PUSH E TOS <- E MUL TOS <- (D*E) SUB TOS <- (C-D*E) PUSH F TOS <- F DIV TOS <- (C-D*E)/F PUSH G TOS <- G DIV TOS <- (C-D*E)/F/G MUL TOS <- (A-B) * (C-D*E)/F/G POP X M[X] <- TOS	**4**	**2**	**2**	**2**	**2.1.2**
12	**Give the IEEE standard for floating point numbers for single and Double precision number.** **Single Precision:** Single Precision is a format proposed by IEEE for the representation of floating-point numbers. It occupies 32 bits in computer memory.	**4**	**2**	**2**	**1**	**2.1.2**

	 **Double Precision:** Double Precision is also a format given by IEEE for the representation of the floating-point number. It occupies 64 bits in computer memory. 					
13	**How carry signal is generated based on the two functions? Write down the expression of carry, based on the propagation and generation for 4-bit operation.** Using the Gi and Pi terms the Sum Si and Carry Ci+1 are given as below – Carry propagated Pi is associated with the propagation of carry from Ci to Ci+1. It is calculated as Pi = Ai ⊕ Bi. Carry generate Gi depends on Ai and Bi inputs. Gi is 1 when both Ai and Bi are 1. Hence, Gi is calculated as Gi = Ai. Bi. • Si = Pi ⊕ Gi. • Ci+1 = Ci.Pi +Gi. Therefore, the carry bits C1, C2, C3, and C4 can be calculated as C1 = C0.P0+G0. C2 = C1.P1+G1 = (C0.P0+G0).P1+G1. C3 = C2.P2+G2 = (C1.P1+G1).P2+G2. C4 = C3.P3+G3 = C0.P0.P1.P2.P3 + P3.P2.P1.G0 + P3.P2.G1 + G2.P3 + G3.	**4**	**2**	**3**	**1**	**2.1.2**
14	**Identify and draw a 4-bit ALU operation of both adder / subtractor circuit using full adder.** The Binary Adder-Subtractor is a combination of 4 Full-Adder, which is able to perform the addition and subtraction of 4-bit binary numbers. The control line determines whether the operation being performed is either subtraction or addition. This determination is done by the binary values 0 and 1, which is hold by K.	**4**	**1**	**3**	**1**	**2.1.2**

						
15	**Give the booth's recoding and bit-pair recoding multiplier for the given number. 1110110010110100 1101** Step1: Add 0 in LSb 1 1 1 0 1 1 0 0 1 0 1 1 0 1 0 0 1 1 0 1 0 Booth Multiplier Recoding 0 0 -1 +1 0 -1 0 +1 -1 +1 0 -1 +1 -1 0 +1 0 -1 +1 -1 Bit Pair Recoding 0 0 -1 +1 0 -1 0 +1 -1 +1 0 -1 +1 -1 0 +1 0 -1 +1 -1 0 -1 -1 +1 -1 -1 +1 +1 -1 +1	**4**	**3**	**3**	**1**	**2.1.2**
	Part – C **(2*12 = 24 Marks)**					
16	**a. Consider AX = 1212h; BX= 1B1Bh; CX=2323h; DX=2C2Ch; SI= 0050h; DI = 0110h; SP= 23h; BP= 0100h; Calculate the effective address and find the addressing modes of the following instructions:** **(i) MOV AX, [2000h] (3Marks)** EA = 2000h Direct Addressing Mode **(ii) MOV AX, 20h [BP] (3Marks)** EA = [BP] + Disp = 0100 + 20 = 0120h Relative Addressing Mode **(iii) MOV AX, 40h [BX][DI] (3Marks)** EA = [BX]+[DI] + Disp = 1B1B+ 0110 + 40 = 1C6Bh Based Indexed Relative Addressing Mode **(iv) MOV AX, 500h (3Marks)** EA = None Immediate Addressing Mode **(OR)** **b. Construct a fast addition process for the below-mentioned binary numbers 11010, 10110, 11011, 10110, 11101, 11111 by converting a three-operand addition to a two-operand.**	**12** **12**	**1** **2**	**2** **3**	**2** **1**	**2.1.2** **2.1.2**

	16b A B C D E F 11010, 10110, 11011, 10110, 11101, 11111 First 3-operand Addition A,B,C & D,E,F  $G_1 + G_2 + G_3 + G_4 + G_5$ 2m + 2m + 3m + 3m + 2m = 12 marks.					
17	**a. Show the multiplication process using Booth's algorithm when the following numbers are multiplied: (-13) by (+8)** **Each Cycle carries 2 marks 5 * 2 = 10 marks** **Final 2's Complement result = 2 marks**  	12 12	3 3	3 3	1 1	2.1.2 2.1.2

17a) $-13 \times +8 = -104$.

Multiplicand in binary $-13 = 11101$

Multiplier in binary $+8 = 01000 \Rightarrow Q$.

1's complent of multiplicand $= 10010 + 1$

2's " " " $= 10011 \Rightarrow M$

$-M = 01100 + 1 = 01101$

A	Q	Q-1	Operation	
00000	01000	0	Initial step	
00000	00100	0	AShr	cycle1
00000	00010	0	AShr	cycle2
00000	00001	0	AShr	cycle3
00000 + 01101 = 01101	00001	0	A=A-M	cycle4
00110	10000	1	AShr	
10011 + 11001	10000	1	A=A+M	cycle5
11100	11000	0	AShr	

The result is in 2's complement form.

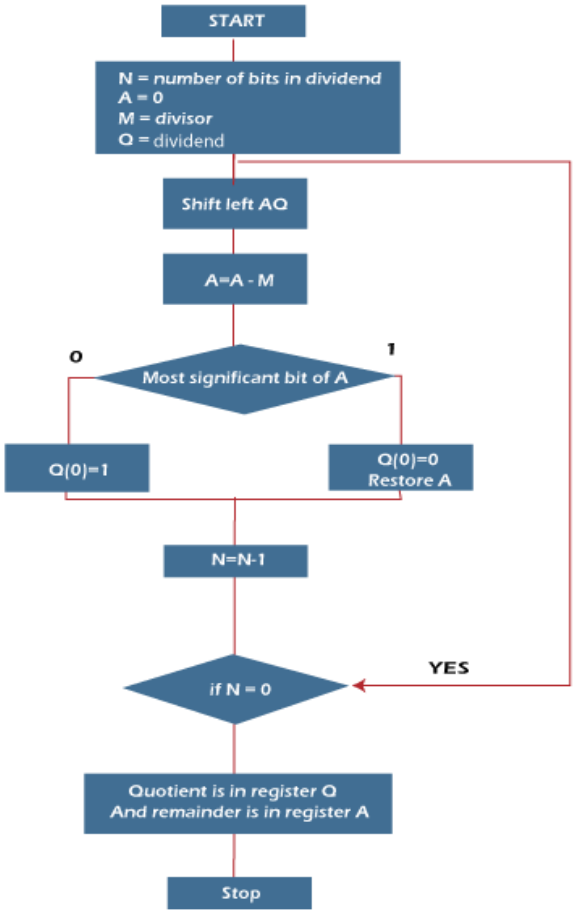
The sign value represent as 1 so the result is negative.

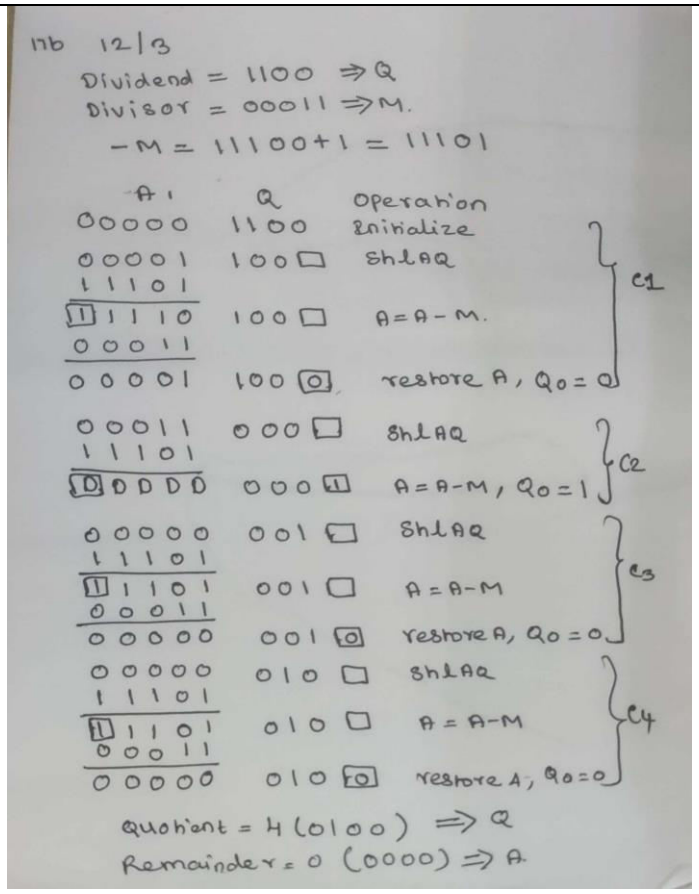
$-00110001111+1 = 001101000$

$= -104$.

(OR)

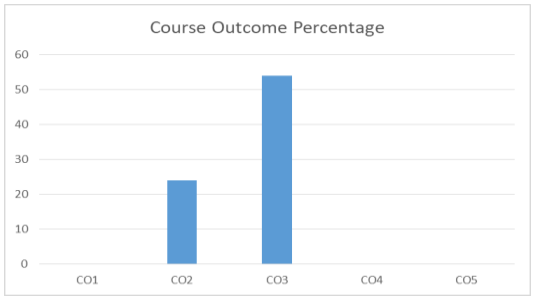
b. Divide $(12)_{10}$ by $(3)_{10}$ using the Restoring division algorithm with step by step intermediate results.

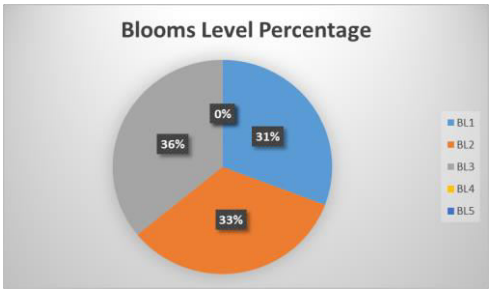


	 Each Cycle Carries 3 marks 4 *3 = 12 marks					

***Performance Indicators are available separately for Computer Science and Engineering in AICTE examination reforms policy.**

Course Outcome (CO) and Bloom's level (BL) Coverage in Questions





Approved by the Audit Professor/Course Coordinator